Stage-Resolved Transcriptomic Analysis Reveals Distinct Molecular Programs Associated with Liver Fibrosis Severity

Abstract

Liver fibrosis severity is associated with heterogeneous transcriptional changes whose consistency and biological organisation across cohorts remain incompletely characterised. We integrated eight public human liver transcriptomic datasets comprising 1,085 samples from 1,027 recorded patient identifiers with curated knowledge-graph evidence. Cross-study analyses identified 3,845 fibrosis-associated genes and 4,199 genes associated with ordered disease-group labels, including 1,822 shared genes. Fibrosis-stage profiles across F0–F4 formed five patterns, including early and late increases, an early rise followed by a plateau, and intermediate- and late-stage decreases. Functional enrichment highlighted extracellular-matrix organisation, collagen remodelling, adhesion, immune processes and metabolic programmes. A score combining association magnitude, endpoint separation, adjacent-stage evidence, operational fibrosis-only status and functional annotations identified 447 Tier 1 genes and 23 cluster-representative candidates. Of the Tier 1 genes, 442 retained statistical significance across all five leave-one-study-out refits. Knowledge-graph analysis contextualised prioritised genes through interconnected biological annotations. These findings provide a framework for investigating transcriptional heterogeneity associated with fibrosis severity. The cross-sectional associations and internal robustness assessments require independent longitudinal and functional validation.

Introduction

Advanced liver fibrosis is associated with liver-related complications and mortality in fatty liver disease [1, 2]. Current terminology distinguishes metabolic dysfunction-associated steatotic liver disease (MASLD) and steatohepatitis (MASH) [3]. Fibrosis may progress, remain stable or regress [4], motivating investigation of severity-associated molecular heterogeneity.

Previous transcriptomic studies linked expression to histological severity (Hoang et al.) [11] and independently replicated a fibrosing-steatohepatitis signature (Govaere et al.) [5]. Cross-cohort studies examined MASLD/MASH signatures and coexpression (Piras and DiStefano) [6], shared and aetiology-specific fibrogenic programmes (Yang et al.) [7], and advanced (F3–F4) versus mild or absent fibrosis (F0–F1), including cellular context (Maiers et al.) [8]. Knowledge graphs such as Hetionet and OptimusKG connect heterogeneous biological evidence across molecular and broader biomedical domains [9, 10], supporting interpretation without independently establishing causality or candidate validity.

The opportunity is to combine cross-study directional agreement, individual F0–F4 gene profiles, adjacent-stage contrasts and curated annotations for transparent prioritisation. This requires harmonising identifiers, expression units and clinical labels, handling repeated biopsies consistently, and assessing unequal stage sizes and cohort composition. Gene stage-profile clustering differs from patient clustering or coexpression analysis; cross-sectional patterns cannot establish within-patient progression.

We aimed to identify reproducible stage-associated patterns and their biological organisation, asking which fibrosis-stage and disease-group associations agree across studies, which profiles characterise F0–F4, and how statistical and annotation evidence can prioritise candidates. Eight datasets were integrated with OptimusKG and drug data from ChEMBL, DGIdb, GtoPdb and Open Targets; drug annotations did not contribute to scoring. The principal contributions were:

- An integrated resource of 1,085 samples, 610 patient-level fibrosis observations and 216 disease-group observations, comprising 126,763 nodes and 35,138,047 relationships.
- Identification of 3,845 fibrosis-associated and 4,199 disease-group-associated genes (1,822 shared), organised into five fibrosis-stage patterns.

- A defined scoring framework identifying 447 Tier 1 genes and 23 candidates; 442 Tier 1 genes retained significance across five leave-one-study-out refits.

These contributions support association, prioritisation and internal robustness; independent validation remains necessary. Results and Discussion precede Materials and Methods.

Results

Integrated resource and cross-study gene associations

Eight datasets comprised 1,085 samples from 1,027 recorded patient identifiers. Repeat-biopsy handling yielded 610 fibrosis observations across five studies and 216 disease-group observations across three. Harmonisation retained 53,993 genes, including 14,794 shared across all datasets. Integration with OptimusKG and ChEMBL, DGIdb, GtoPdb and Open Targets produced 126,763 nodes and 35,138,047 relationships, including 12,544 Drug nodes and 25,890 drug–gene links verified by direct queries. Drug annotations were excluded from transcriptomic analyses.

Of 3,845 fibrosis-associated and 4,199 disease-group-associated genes, 1,822 were shared, 2,023 fibrosis-only and 2,377 disease-group-only (Fig. 1a). Shared genes included 1,408 positive and 384 negative associations on both axes, with 30 directionally discordant (Fig. 1b). These categories reflect statistical selection across different populations, not established biological specificity.

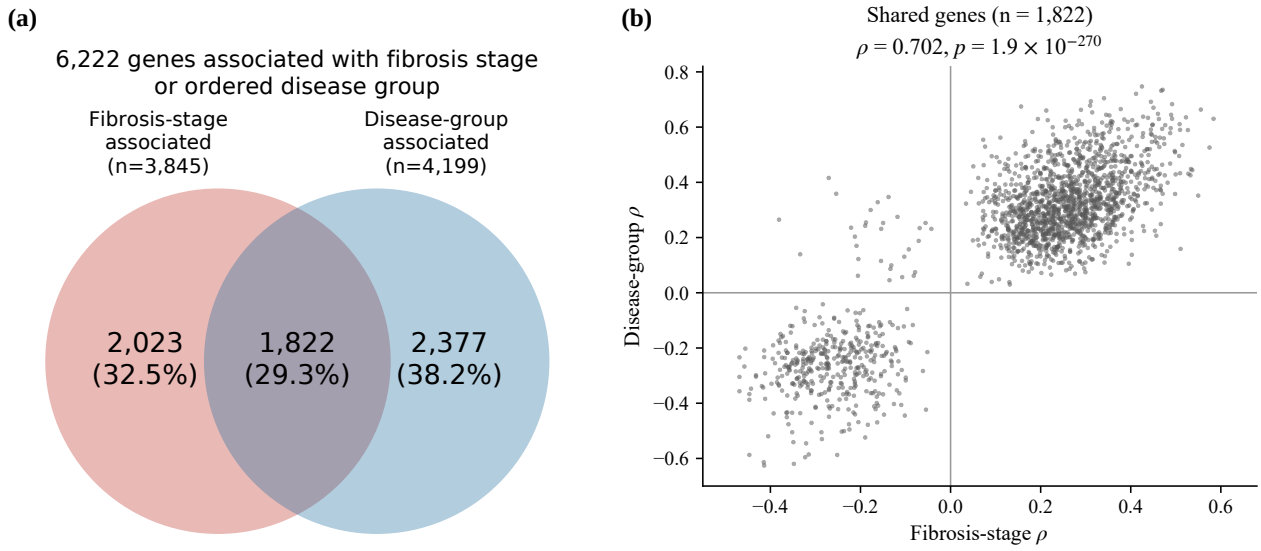


Figure 1: Fibrosis and disease-group associations. (a) Gene-set overlap. (b) Median study correlations of the 1,822 shared genes (Spearman across genes). Selection: BH $q < 0.05$, consistent direction in all testable studies, $\geq$3 studies. Disease groups (control/NAFL/NASH) are ordered labels, not inflammation measures.

Five expression patterns characterised fibrosis severity

The fibrosis-associated genes formed five clusters with distinct expression profiles across F0–F4 (Fig. 2). C1 contained 857 genes showing an early increase followed by continued elevation. C2 comprised 1,367 genes with a late increase, whereas C3 contained 226 genes showing an early rise followed by a plateau. Decreasing profiles comprised C4, with 397 genes showing a mid-stage decrease, and C5, with 998 genes showing a late decrease.

The five-cluster solution satisfied the predefined gene-subset stability criterion. Mean adjusted Rand index was 0.983 for gene-subset refits, compared with 0.592 for patient-bootstrap refits. Although average patterns

were reproducible, individual assignments were less stable: 1,838 genes met the membership-confidence threshold of 0.8. The expression-profile heatmap is provided in Supplementary Fig. S1, and cluster-number diagnostics are shown in Supplementary Fig. S2 and Table S1; individual membership results accompany the analysis outputs. These patterns represent cross-sectional differences between patients at different stages, rather than observed trajectories within individuals.

Table 1: Cross-sectional clusters, confidence cores and prioritisation

Cluster	Pattern	Genes	Confidence core≥ 0.8	Tier1	Shortlist	Mean F0	Mean F4
C1	Early increase (F0→F1), then continued	857	90	93	5	-0.295	0.515
C2	Late increase (largest at F3→F4)	1367	806	262	5	-0.227	0.86
C3	Early rise, then plateau	226	74	7	5	-0.27	0.136
C4	Mid decrease (largest at F1→F2)	397	126	3	3	0.161	-0.279
C5	Late decrease (largest at F3→F4)	998	742	82	5	0.184	-0.658

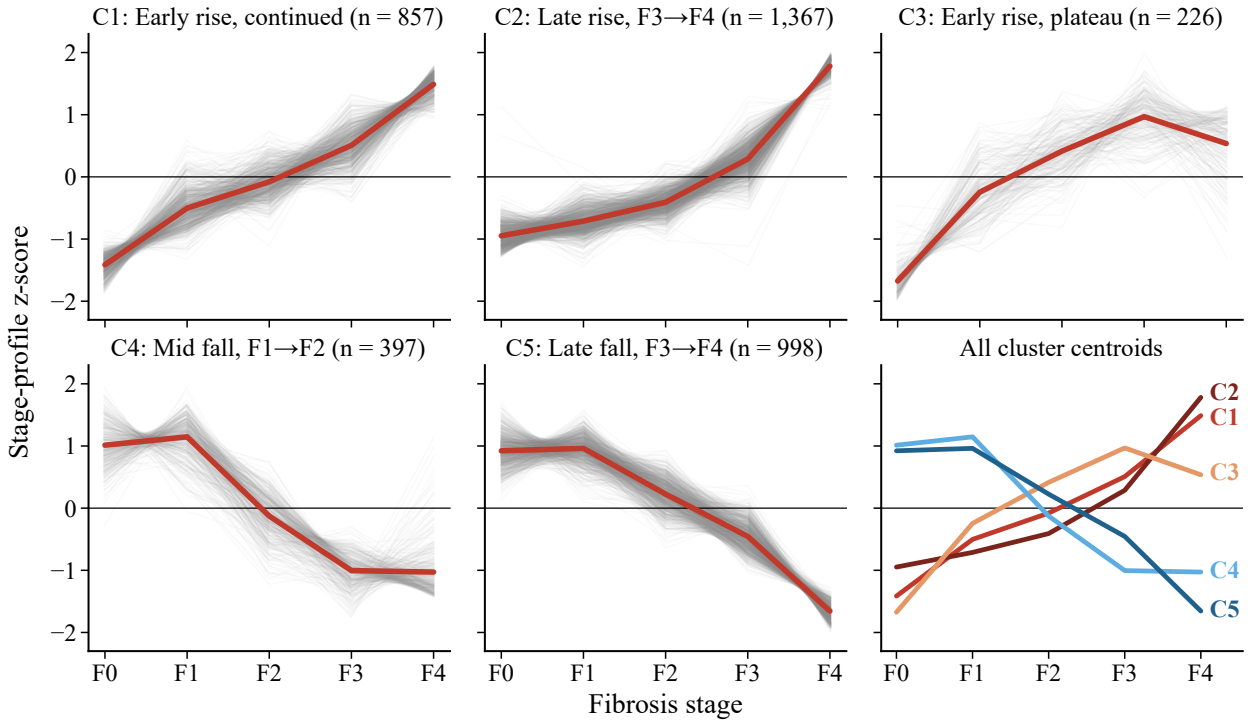
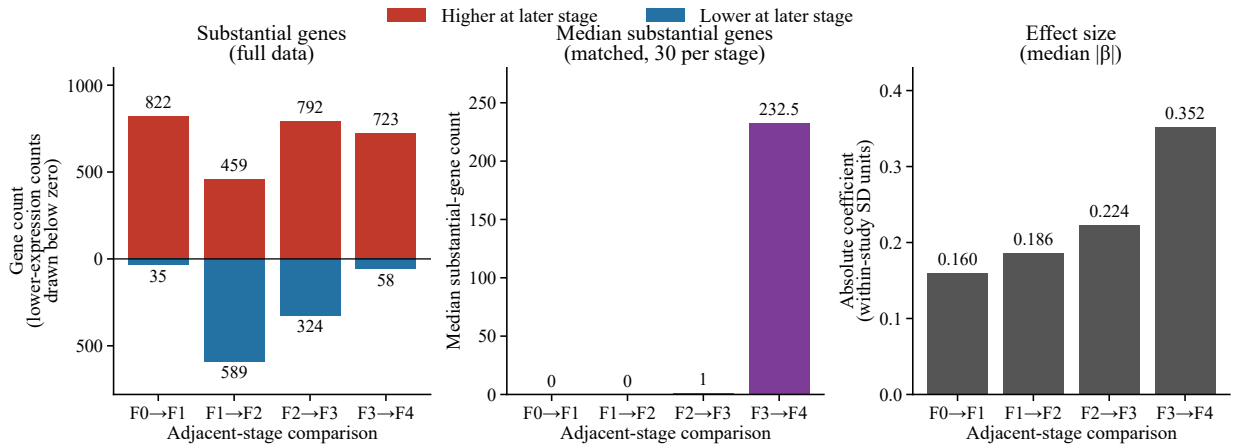


Figure 2: Fibrosis-stage expression patterns. Individual gene profiles (grey) and cluster centroids (red), standardised across F0–F4. Profiles describe cross-sectional stage differences, not longitudinal trajectories; grey lines represent genes, not confidence intervals. The corresponding heatmap and clustering diagnostics are provided in Supplementary Figs. S1 and S2.

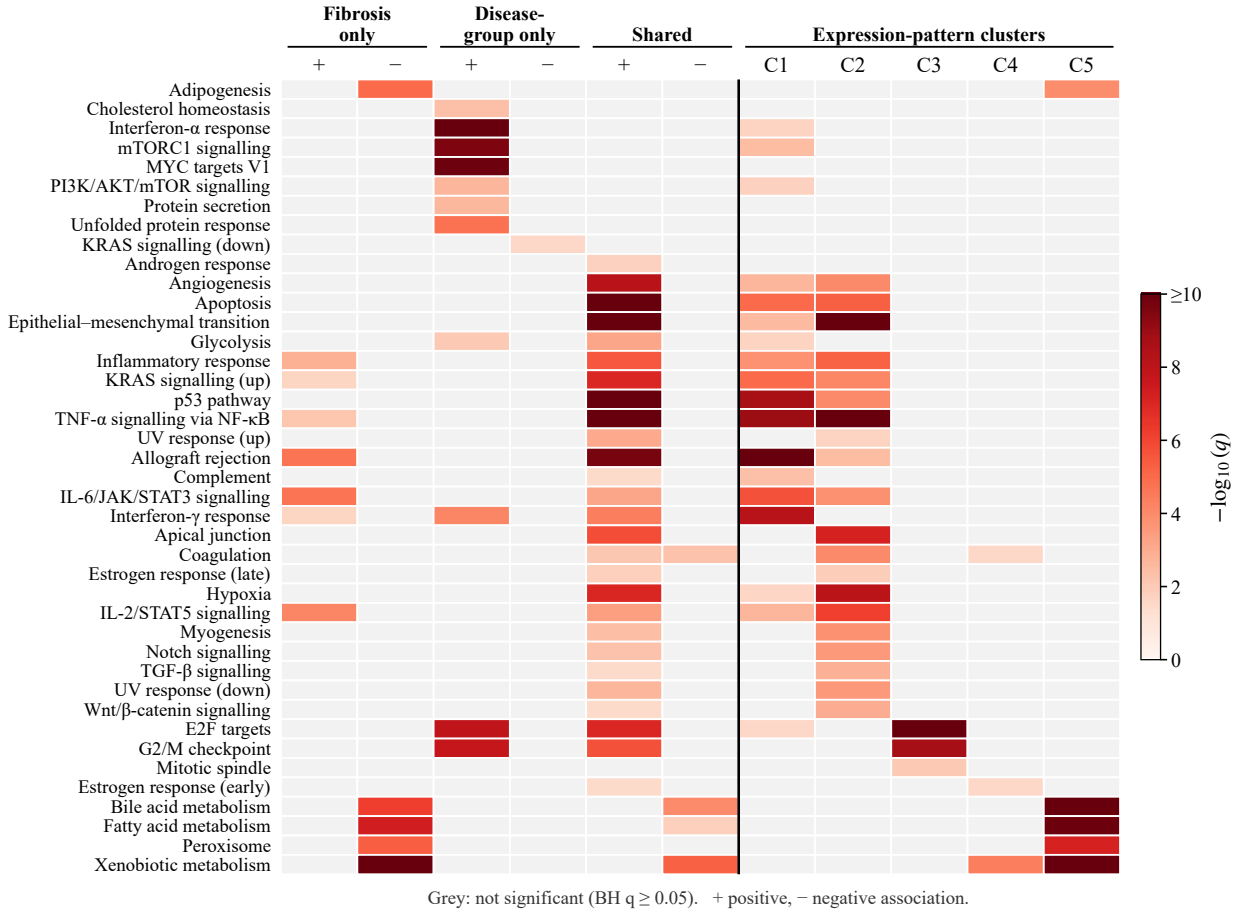
Adjacent-stage contrasts and enrichment revealed distinct biological contexts

Study-adjusted comparisons identified 857 substantial expression differences between F0–F1, 1,048 between F1–F2, 1,116 between F2–F3 and 781 between F3–F4, using $q < 0.05$ and $|\beta| \geq 0.2$ (Fig. 3a). F2–F3 had the largest number of substantial differences, while F3–F4 had the largest median absolute coefficient, at 0.352 within-study standard-deviation units. However, F3–F4 included only 34 F4 observations. Sample-size-matched refits produced markedly different counts and broad distributions, indicating sensitivity to patient composition and statistical power.

Functional enrichment distinguished immune, matrix-associated and metabolic annotations (Fig. 3b). Fibrosis-only increasing genes were enriched for T-cell activation and IL6–JAK–STAT3 signalling, while



(a)



(b)

Figure 3: Adjacent-stage contrasts and functional enrichment. (a) Substantial-gene counts, matched-refit median counts and median absolute study-adjusted coefficients. Substantial differences require BH $q < 0.05$ and $|\beta| \geq 0.2$ SD units; matched refits use 30 patients per stage. (b) Hallmark enrichment across gene groups and clusters (14,794-gene background). Colour denotes $-\log_{10}(q)$, capped at 10.

genes positively associated with both clinical axes were enriched for collagen formation and extracellular matrix organisation. The late-increasing C2 cluster was enriched for the Hallmark epithelial–mesenchymal transition set ($q = 5.25 \times 10^{-27}$); C3 was enriched for E2F targets ($q = 1.24 \times 10^{-14}$). The late-decreasing C5 cluster was enriched for bile-acid and fatty-acid metabolism.

Enrichment depended partly on cluster-membership confidence, particularly for C4, whose full-cluster and confidence-core analyses shared no significant Reactome term. Complete enrichment results accompany the

analysis outputs. These findings indicate annotation over-representation rather than demonstrated activation or suppression of biological processes.

Multi-evidence prioritisation identified 23 representative candidates

The prioritisation framework identified 447 Tier 1 genes: 93 in C1, 262 in C2, seven in C3, three in C4 and 82 in C5. Selecting up to five qualifying genes per cluster yielded 23 candidates because C4 contained only three genes meeting the threshold (Table 2). PDGFRA, CACNA1C, TPM4, PLK3 and TAX1BP3 achieved the maximum score of nine. The complete candidate table is shown in Table 2.

Table 2: Final 23-gene shortlist

Gene	Cluster	Score /9	Median fibrosis rho	Membership confidence	Valid bootstrap denominator	Largest absolute contrast
IL32	C1	8	0.522	0.83	100	F0→F1
EPHA3	C1	8	0.498	0.46	100	F2→F3
LYZ	C1	8	0.487	0.83	100	F0→F1
NFASC	C1	8	0.532	0.54	100	F2→F3
PLPP4	C1	8	0.533	0.58	100	F2→F3
PDGFRA	C2	9	0.417	0.68	97	F2→F3
CACNA1C	C2	9	0.447	0.89	97	F3→F4
TPM4	C2	9	0.403	0.89	97	F3→F4
PLK3	C2	9	0.414	0.89	97	F3→F4
TAX1BP3	C2	9	0.408	0.92	97	F3→F4
TREM2	C3	8	0.469	0.67	99	F0→F1
CDC45	C3	7	0.401	0.75	99	F0→F1
HTR7	C3	7	0.44	0.71	99	F2→F3
FABP5	C3	7	0.464	0.73	99	F0→F1
TYMS	C3	7	0.489	0.61	99	F0→F1
OAT	C4	7	-0.398	0.38	100	F2→F3
KNG1	C4	7	-0.312	0.71	100	F1→F2
F2	C4	7	-0.307	0.56	100	F1→F2
SULT1E1	C5	8	-0.414	0.99	96	F3→F4
SLC38A4	C5	8	-0.43	0.99	96	F3→F4
CYP2C19	C5	8	-0.443	0.76	96	F1→F2
ACADSB	C5	8	-0.447	0.89	96	F1→F2
MME	C5	8	-0.421	0.99	96	F3→F4

Leave-one-study-out analysis retained statistical significance for 442 of the 447 Tier 1 genes across all five refits, while 247 retained an absolute median correlation of at least 0.30 throughout. All 23 shortlisted genes remained significant, and 18 retained the magnitude threshold. Nevertheless, shortlist membership confidence ranged from 0.38 to 0.99, indicating that high prioritisation scores did not ensure equally stable cluster assignments.

Sixteen candidates were retained across all four label/transformation configurations. Removing the fibrosis-only component while retaining the score threshold reduced the Tier 1 set to 286 genes and the shortlist to 21; this comparison also reduced the maximum achievable score. These analyses assess internal robustness and sensitivity, rather than independent validation.

Knowledge-graph annotations contextualised prioritised genes

The filtered annotation network connected 42 Tier 1 genes to twelve selected pathway or biological-process annotations through 154 relationships. Enriched annotations included extracellular matrix organisation, cell–cell adhesion, angiogenesis, integrin interactions and PDGF signalling.

Four shortlisted genes appeared in this network: PDGFRA connected to five selected annotations, while EPHA3, NFASC and TREM2 each connected to one. The remaining displayed genes belonged to the wider

Tier 1 set and each reached at least three annotations. The network therefore represents annotation sharing within a selected subset, rather than convergence across the complete shortlist. The annotation subgraph is shown in Supplementary Fig. S3; complete enrichment statistics accompany the analysis outputs.

Discussion

This study identified distinct transcriptional patterns associated with liver fibrosis severity and organised their statistical and biological context within an integrated knowledge graph. By combining cross-study directional agreement, F0–F4 expression profiles, adjacent-stage contrasts and curated annotations, the framework complements studies linking liver expression to histological severity [11]. Its contribution is an interpretable organisation of evidence for candidate investigation; clinical prediction and superiority over existing methods were not evaluated.

Increasing expression patterns were associated with extracellular-matrix, collagen, adhesion and immune annotations, whereas the late-decreasing C5 pattern was enriched for bile-acid and fatty-acid metabolism. These findings suggest differences in tissue-remodelling and metabolic programmes across fibrosis stages. However, enrichment does not establish pathway activation or causal mechanisms, and bulk-liver expression cannot distinguish changes within cells from changes in cellular composition. The shortlisted TREM2 and PDGFRA genes are compatible with independent observations of scar-associated macrophages and mesenchymal populations in human cirrhosis [23], supporting cell-resolved investigation without establishing their cellular origin in these cohorts. Likewise, developmental annotations indicate shared gene membership rather than direct evidence of the named process in liver tissue.

The five-pattern representation was stable under gene subsampling but sensitive to patient composition (mean ARI 0.983 versus 0.592). Only 1,838 genes met the conditional membership-confidence threshold. Because the highest silhouette score occurred at $k = 3$, the five clusters represent the stated stability-based resolution rather than a uniquely established biological partition. Individual cluster assignments should therefore be interpreted alongside membership confidence.

Adjacent-stage comparisons provided local expression contrasts, but unequal stage sizes and variable matched-refit results limit claims of a discrete biological transition. F3–F4 had the largest median absolute coefficient, yet included only 34 F4 observations. Study adjustment did not eliminate clinical or cell-composition confounding. Most importantly, these cross-sectional profiles describe differences between patients and cannot establish molecular progression within individuals.

The 23-gene shortlist represents all five patterns without relaxing qualification thresholds for smaller clusters. LOSO significance retention was high among Tier 1 genes, but fewer retained the effect-magnitude criterion across every refit (442 versus 247); the corresponding shortlist counts were 23 and 18. Statistical significance and effect stability therefore provide complementary measures of internal robustness, rather than independent candidate validation. Score components also share evidence, and sensitivity to label, transformation and scoring choices remains relevant. For example, removing the fibrosis-only component reduced Tier 1 membership to 286 genes while retaining the seven-point threshold under a reduced maximum score.

Graph connectivity contextualised a subset of the shortlist: PDGFRA reached five selected annotations, whereas EPHA3, NFASC and TREM2 reached one each. This does not imply convergence of all candidates on the displayed network. Similarly, fibrosis-only status reflects statistical selection in different analysis populations, not established biological specificity. Curated annotation coverage and reuse of annotation evidence in prioritisation constrain interpretation of graph connectivity as corroboration.

Dataset selection was not systematic, and participant overlap across accessions was unresolved. The 73 retained GSE130970 disease classifications were histology-derived proxies rather than recovered original diagnoses. Incomplete metabolic and alcohol information prevented reconstruction of current MASLD/MASH criteria [3]; age and sex were available for only 327 of 610 staged patients. These limitations, unequal stage representation and heterogeneous source measurements restrict generalisability and confounder control.

Input manifests, checksums and explicit parameters support reproducibility, but the retrospectively generated manifest does not document every acquisition decision or establish official HGNC/OptimusKG release dates. The common expression core also does not constitute a complete graph-measurement archive. Drug-annotation coverage varied and therapeutic efficacy was not assessed. Candidate selection was not nested within an independent evaluation design. Independent-cohort replication, longitudinal sampling and functional experiments are therefore needed to establish candidate generalisability, temporal relevance and biological roles.

Materials and Methods

Study design and data sources

We conducted a retrospective computational study of public human liver transcriptomic data to identify fibrosis-severity associations, characterise F0–F4 expression patterns and prioritise candidate genes using curated biological evidence (Fig. 4). Eight GEO datasets [12] comprised 1,085 samples from 1,027 recorded patient identifiers (Table 3) [13, 11, 5, 14, 15, 16, 17, 18]. Identifiers were interpreted within studies; participant independence across accessions was not established.

These human liver tissue datasets provided sample-level expression files and linkable metadata. They were not selected through a documented systematic search. Deposited count, TPM, GCT and per-sample expression files were analysed without new sequencing, alignment or quantification. Clinical annotation availability determined eligibility for each analysis (Supplementary Table S2).

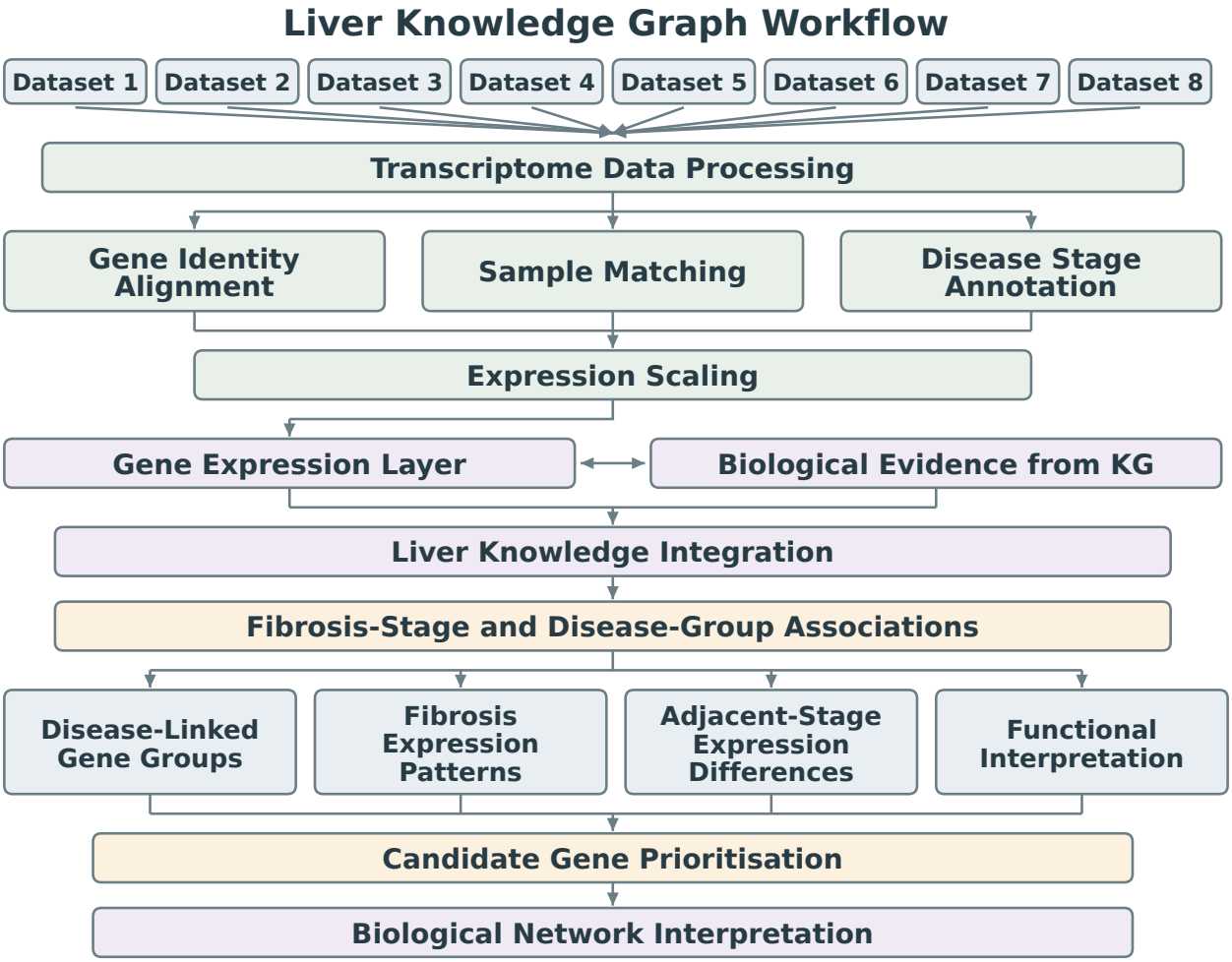


Figure 4: Study workflow. Processing and harmonisation of eight liver transcriptomic datasets, knowledge-graph integration, cross-study association analysis, stage-profile clustering, functional enrichment and candidate-gene prioritisation.

Table 3: Public liver transcriptomic datasets and clinical annotation coverage

Accession	Ref.	Samples	Recorded patient IDs	Staged biopsies	Samples with disease labels	Deposited expression file
GSE126848	[13]	57	57	0	57	GSE126848_Gene_counts_raw.txt.gz
GSE130970	[11]	78	78	78	73	GSE130970_all_sample_salmon_ tximport_TPM_entrez_gene_ID.csv.gz
GSE135251	[5]	216	216	216	216	GSE135251_RAW.tar
GSE162694	[14]	143	143	143	143	GSE162694_raw_counts.csv.gz
GSE167523	[15]	98	98	0	98	GSE167523_Raw_gene_counts_matrix.txt.gz
GSE193066	[16]	164	106	164	0	GSE193066_NAFLD.HUn164.gct.gz
GSE240729	[17]	67	67	67	0	GSE240729_merged.counts.67samp.txt.gz
GSE269412	[18]	262	262	0	0	GSE269412_DRX0FLO_rnaseq_count_ mat.txt.gz
Total		**1,085**	**1,027**	**668**	**587**	—

Counts describe the resource before repeat-biopsy resolution. Disease-labelled samples include broad categories excluded from ordered analysis; five GSE130970 cases are unclassified. Final eligible counts are given in Supplementary Table S2.

Clinical annotation and patient handling

Expression samples were linked to GEO records using study-specific identifiers, titles and descriptions. Harmonised fields included patient identity, biopsy order, fibrosis stage and disease group; original fields and provenance were retained. Fibrosis stages were ordered F0–F4, and disease groups were ordered control = 0, NAFL = 1 and NASH = 2. Broad NAFLD, obese-only and unclassified labels were excluded from the disease-group analysis. Original diagnoses were not assumed equivalent to contemporary MASLD/MASH criteria [3].

GSE130970 lacked recovered individual author-assigned diagnoses. Operational classifications used histological components: zero steatosis defined a control proxy; positive steatosis with positive ballooning defined a NASH proxy; and positive steatosis with zero ballooning defined a NAFL proxy [11]. Five ambiguous classifications were set to unclassified, retaining their fibrosis stages; 73 histology-derived labels remained. These proxies do not constitute complete histopathological diagnoses.

For repeated biopsies, standardised expression was averaged when all recorded stages were known and identical; otherwise, only the first biopsy was retained. Selection preceded stage-specific filtering and was applied throughout discovery, clustering, comparisons and resampling. In GSE193066, 28 same-stage pairs were averaged and 30 different-stage pairs contributed their first biopsy, resolving 164 biopsies to 106 patient observations [16]. Studies required at least 25 patient observations and variation on the clinical axis. The fibrosis analysis included 610 observations across five studies; the disease-group analysis included 45 from GSE126848 (excluding 12 obese-only samples), 73 from GSE130970 and 98 from GSE167523 (216 total; Supplementary Table S2).

Gene harmonisation and expression preprocessing

Missing and non-numeric measurements were removed, measured zeros were retained, and duplicate raw identifiers were resolved by keeping their first occurrence. The HGNC complete set mapped approved symbols, Entrez identifiers and unambiguous aliases or previous symbols to Ensembl identifiers. Ensembl version suffixes and _PAR_Y rows were removed; Ensembl identifiers without HGNC annotation were retained, and mapping failures were recorded. Distinct raw identifiers mapping to the same sample–gene pair were summed.

GSE130970 was transformed as $\log_2(\text{TPM} + 1)$; count datasets used $\log_2(\text{CPM} + 1)$, with library totals calculated before mapping and filtering. GSE193066 retained its deposited linear RLE-normalised counts and used $\log_2(\text{count} + 1)$ without CPM rescaling [19]. No DESeq2 differential-expression model was fitted. Mapped-gene summation preceded transformation, followed by within-study gene standardisation (sample SD, ddof $= 1$), then patient-level selection or averaging without restandardisation. Undefined SDs yielded zero standardised expression. GSE193066 standardisation included all 164 biopsies before patient aggregation. No fitted cross-study batch correction was applied. Effects are in within-study SD units, not fold changes. The graph retained the measured-gene union; association eligibility used the 14,794 genes measured in all eight studies. Dataset-specific processing is detailed in Supplementary Table S2.

Cross-study association analysis

Within each eligible study, Spearman correlation related patient-level standardised expression to each ordered clinical axis, using average ranks for ties. Two-sided p-values used $t = \rho\sqrt{(n-2)/(1-\rho^2)}$ with $n-2$ degrees of freedom; correlations were clipped to ± 0.9999999. Testability required complete expression coverage and nonconstant ranked expression. Genes required at least three testable studies. The median correlation summarised direction and magnitude; Fisher's statistic combined two-sided p-values independently of direction:

$$Q_g = -2\sum_{d=1}^{D_g} \log(p_{gd}), \qquad Q_g \sim \chi^2_{2D_g}.$$

Here, D_g denotes the number of testable studies, and p-values were bounded below by 10^{-300}.

BH adjustment [20] preceded directional filtering and was applied separately to 14,794 fibrosis tests and 14,784 disease-group tests. Ten common genes were constant among the 45 eligible GSE126848 observations and failed the three-study requirement. Selection required $q < 0.05$ and every testable study correlation to share the median's positive/non-positive class; no minimum correlation magnitude was imposed. Set membership defined fibrosis-only, disease-group-only and shared genes, with shared genes separated by direction agreement. These categories describe selection in different populations rather than biological specificity. Exported stage means and median correlations were rounded to three decimal places for downstream summaries and scoring.

Fibrosis-stage expression profiles and clustering

Mean patient-level expression at each stage formed a five-value profile for each selected fibrosis-associated gene. Profiles were row-standardised (ddof $= 0$) and clustered by K-means (50 initialisations; seed 42), emphasising shape. Candidate $k = 3$–8 solutions were evaluated by silhouette scores and ARI stability across 50 refits using 80% gene subsets and a separate 50 within-study patient bootstraps (10 initialisations per diagnostic fit). The largest k with mean gene-subset ARI ≥ 0.95 was selected. Patient-bootstrap stability was assessed separately (Supplementary Table S1).

A further 100 patient bootstraps assessed membership confidence. Hungarian assignment matched bootstrap and reference centroids by correlation. A gene was assessable only when its reference cluster's matched centroid had $r \geq 0.5$; confidence was the fraction of assessable refits returning it to that cluster, with denominators retained. Confidence-core membership required ≥ 0.8. As a sensitivity check, profiles estimated by ordinary least squares with categorical stage and study terms were row-standardised, reclustered and compared by ARI and per-gene correlations. Cluster names describe cross-sectional patterns, not individual longitudinal trajectories.

Adjacent-stage comparisons and matched refits

F0–F1, F1–F2, F2–F3 and F3–F4 comparisons used selected fibrosis-associated genes and studies represented at both stages. Ordinary least squares estimated the higher-stage effect β_g with study-specific intercepts:

$$z_{ig} = \alpha_g + \beta_g H_i + \sum_{d=2}^{D} \gamma_{gd} I(\text{study}_i = d) + \varepsilon_{ig}.$$

Two-sided t tests assessed β_g, with BH correction across the selected fibrosis genes within each contrast. Substantial differences required $q < 0.05$ and $|\beta_g| \geq 0.2$ SD units.

To assess unequal stage sizes, 50 matched draws (seed 42) sampled without replacement within study–stage pools: two patients per stage from GSE130970, 14 from GSE135251, eight from GSE162694 and six from GSE240729 (30 per stage). Reproducibility was the fraction of draws with $|\beta| \geq 0.2$ and the full-data coefficient's sign; significance was recorded separately. Analyses remained conditional on fibrosis-gene selection, and study adjustment did not comprehensively control clinical or cell-composition differences.

Functional enrichment analysis

GSEApy [22] performed one-sided hypergeometric over-representation tests using MSigDB 2024.1.Hs GO Biological Process, KEGG legacy, Reactome and Hallmark collections [21]. The background contained 14,794 unique harmonised symbols corresponding one-to-one to the common Ensembl eligibility universe. Annotation sets were intersected with this background and restricted to 10–500 genes; input groups required at least five genes. BH adjustment was applied within each group and collection ($q < 0.05$).

Tests covered increasing and decreasing fibrosis-only, disease-group-only and concordant shared groups, five full clusters and five confidence-core sets. Discordant shared genes were excluded from the shared direction groups. Outputs included overlap genes, counts, fold enrichment and adjusted p-values; full-cluster/core comparisons assessed membership sensitivity. Undersized groups were marked untested. Enrichment describes annotation over-representation rather than demonstrated pathway activation.

Gene prioritisation and robustness assessment

Fibrosis-associated genes received the nine-point score

$$S_g = P_g + T_g + X_g + F_g + B_g.$$

P_g scored median association magnitude (0–2); T_g, direction-consistent F0–F4 separation (0–2); X_g, the strongest adjacent-stage contrast and matched-refit reproducibility (0–2); F_g, operational fibrosis-only membership (0–1); and B_g, support from enriched annotation collections (0–2). Supplementary Table S3 specifies every threshold, rounding rule, tie-break and missing-evidence rule. Missing selected-contrast evidence stopped scoring rather than assigning zero. Annotation support used all significant terms from the gene's concordant direction group or full cluster, counted each collection once and excluded confidence-core enrichment. Tier 1 required $S_g \geq 7$, Tier 2 required $5 \leq S_g < 7$, and Tier 3 comprised the remainder. Up to five Tier 1 genes per cluster were selected by descending score, then absolute median fibrosis correlation. Correlated score components provide prioritisation rather than independent validation.

LOSO analysis omitted each fibrosis study in turn, repeated patient-level associations and applied BH correction across the 14,794-gene family. Tier 1 summaries separately assessed $q < 0.05$, direction and $|\,\text{median}\,\rho| \geq 0.30$. This assessed internal robustness, not independent held-out prediction.

Sensitivity analyses compared a baseline retaining all 78 original derived GSE130970 classifications and CPM-rescaling GSE193066; labels-only exclusion of the five ambiguous classifications; preservation of deposited GSE193066 RLE normalisation alone; and both changes together. The combined configuration was primary: the five cases were unclassified only for disease-group analysis. Biopsy handling, eligibility, $k = 5$, score thresholds, cluster quotas and annotation selection were fixed. Further checks varied score thresholds, removed individual components and used an alternative cluster-top-score shortlist. The F_g ablation retained the Tier 1 threshold of seven while reducing the maximum score to eight.

Knowledge-graph construction and analysis

Neo4j integrated expression measurements with OptimusKG using Ensembl identifiers [10]. Drug annotations from ChEMBL, DGIdb, GtoPdb and Open Targets enriched the resource but did not contribute to the reported transcriptomic or prioritisation analyses. Eight node categories represented Gene, Sample, Dataset, Disease, Drug, Pathway, BioProcess and Phenotype. EXPRESSES stored raw and standardised sample–gene measurements; IN_DATASET linked samples to studies. Sample properties retained patient identity and biopsy order.

Curated relations represented gene associations and interactions, pathway/process membership, disease hierarchies, phenotypes and drug actions or indications. Open Targets gene–disease scores supplied through OptimusKG required score ≥ 0.1; symmetric interactions were deduplicated. Drug action, indication, contraindication and off-label meanings were retained. TRACKS_FIBROSIS and TRACKS_INFLAMMATION stored selected associations and their statistics; the latter denotes ordered disease-group associations, not direct inflammation measurement. Graph composition, the 14 relationship types and property coverage are described in Supplementary Table S4 and the schema. Association records and displayed subgraph connections were checked against analysis exports.

For a focused annotation network, hypergeometric tests assessed Tier 1 over-representation using the common 14,794-Ensembl-gene background, equivalent to the functional-enrichment symbol background. Annotation sizes counted background-intersecting members; unannotated genes remained in the denominator.

BH correction covered all 3,980 Pathway/BioProcess nodes reached by Tier 1 genes before display filtering. Eligible annotations required 10–500 background genes, at least ten Tier 1 members and $q < 0.05$. Ranked by adjusted p-value, annotations were skipped if their Tier 1 member sets had Jaccard similarity > 0.5 to a retained set; twelve were retained. Display included genes linked to at least three selected annotations or shortlisted genes linked to at least one. This contextual network was not independent validation.

Software and reproducibility

Analyses used Python 3.12.14, pandas 2.3.3, NumPy 2.2.4, SciPy 1.16.3, scikit-learn 1.7.2, matplotlib 3.10.7 and GSEApy 1.3.1; the graph used Neo4j Enterprise 2026.09.0. Clustering and matched refits used seed 42. Code, runtime requirements, tables and provenance manifests document the procedures. The expression-analysis input contains the common 14,794-gene core rather than the complete graph expression layer.

The OptimusKG deposit is identified by DOI 10.7910/DVN/IYNGEV. Formal HGNC/OptimusKG release identifiers and expression-data acquisition dates were not recorded. Retained reference-file modification dates and SHA-256 checksums are documented in the Supplementary Methods and source manifest; these do not establish official release dates. No new human participant data were collected.

Data availability

The source transcriptomic datasets are available from the Gene Expression Omnibus under accessions GSE126848, GSE130970, GSE135251, GSE162694, GSE167523, GSE193066, GSE240729 and GSE269412. The OptimusKG source deposit is identified by DOI 10.7910/DVN/IYNGEV; reuse is subject to the terms of the contributing sources. The reference-file manifest, source_checksums.txt, documents the retained HGNC and OptimusKG files. Drug annotations were obtained from OptimusKG, ChEMBL (REST API, October 2026), DGIdb 5, GtoPdb 2026.3.1 and Open Targets Platform 26.09; source identifiers are retained on the graph relationships (Supplementary Table S4). [AUTHOR_CONFIRM: permanent repository links or DOIs for the final graph exports, processed analysis inputs, complete result tables and supplementary outputs; specify any access or redistribution restrictions.] The shared expression-analysis core covers the common measured-gene universe and is not a complete archive of all graph measurements.

Code availability

[AUTHOR_CONFIRM: public repository URL, immutable release or commit identifier, software licence and archival DOI for the code corresponding to the reported analyses.] The deposited release should include the runtime requirements and instructions linking the source inputs to the analysis outputs and figures.

Ethics approval and consent

This study used publicly available human liver transcriptomic datasets and their associated annotations; no new participant recruitment or sample collection was undertaken. [AUTHOR_CONFIRM: applicable institutional determination for secondary-data ethics review and the consent and data-use conditions governing the source datasets.]

AI-assisted tools

Claude and Codex were used to assist with project support and manuscript preparation. [AUTHOR_CONFIRM: specify the scope of AI-assisted analysis, coding and writing, the tools or versions

where available, and the human verification performed. Confirm that the authors reviewed the final text and outputs and take responsibility for the manuscript.]

Acknowledgements

[AUTHOR_CONFIRM: contributors to acknowledge, the nature of their assistance and permission to name them.]

Funding

[AUTHOR_CONFIRM: funding organisations and grant identifiers, or a verified statement that no specific funding was received; describe the funders' role where applicable.]

Author contributions

[AUTHOR_CONFIRM: author initials and their contributions to conceptualisation, data curation, methodology, software, formal analysis, visualisation, supervision, writing and revision. Confirm approval of the submitted manuscript by all authors.]

Competing interests

[AUTHOR_CONFIRM: financial and non-financial competing-interest declarations for every author.]

Additional information

Correspondence and requests for materials should be addressed to [CORRESPONDING AUTHOR]. Supplementary Table S1 reports cluster-number diagnostics; the Supplementary Methods describe implementation checks and reference-file provenance. [AUTHOR_CONFIRM: permanent location and final inventory of the complete supplementary deposit.]

References

[1] Angulo, P. et al. Liver Fibrosis, but No Other Histologic Features, Is Associated With Long-term Outcomes of Patients With Nonalcoholic Fatty Liver Disease. *Gastroenterology* **149**, 389–397.e10 (2015). https://doi.org/10.1053/j.gastro.2015.04.043.

[2] Sanyal, A. J. et al. Prospective Study of Outcomes in Adults with Nonalcoholic Fatty Liver Disease. *N. Engl. J. Med.* **385**, 1559–1569 (2021). https://doi.org/10.1056/NEJMoa2029349.

[3] Rinella, M. E. et al. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. *Hepatology* **78**, 1966–1986 (2023). https://doi.org/10.1097/HEP.0000000000000520.

[4] Singh, S. et al. Fibrosis progression in nonalcoholic fatty liver vs nonalcoholic steatohepatitis: a systematic review and meta-analysis of paired-biopsy studies. *Clin. Gastroenterol. Hepatol.* **13**, 643–654.e1–9 (2015). https://doi.org/10.1016/j.cgh.2014.04.014.

[5] Govaere, O. et al. Transcriptomic profiling across the nonalcoholic fatty liver disease spectrum reveals gene signatures for steatohepatitis and fibrosis. *Sci. Transl. Med.* **12**, eaba4448 (2020). https://doi.org/10.1126/scitranslmed.aba4448.

[6] Piras, I. S. & DiStefano, J. K. Comprehensive meta-analysis reveals distinct gene expression signatures of MASLD progression. *Life Sci. Alliance* **7**, e202302517 (2024). https://doi.org/10.26508/lsa.202302517.

[7] Yang, W. et al. Integrative Transcriptomics Across Etiologies Reveals Common and Disease-Specific Fibrogenic Signatures in Liver Fibrosis. *Can. J. Gastroenterol. Hepatol.* **2026**, e9254321 (2026). https://doi.org/10.1155/cjgh/9254321.

[8] Maiers, J. L. et al. Conserved hepatic RNA signatures across multiple cohorts reveal novel mechanistic clues for advanced fibrosis in human MASLD. *Hepatol. Commun.* **10**, e0991 (2026). https://doi.org/10.1097/HC9.0000000000000991.

[9] Himmelstein, D. S. et al. Systematic integration of biomedical knowledge prioritizes drugs for repurposing. *eLife* **6**, e26726 (2017). https://doi.org/10.7554/eLife.26726.

[10] Vittor, L. et al. Unifying biomedical knowledge in a modern multimodal graph. *arXiv* 2604.27269v3 (2026), preprint. https://arxiv.org/abs/2604.27269v3; https://doi.org/10.48550/arXiv.2604.27269.

[11] Hoang, S. A. et al. Gene Expression Predicts Histological Severity and Reveals Distinct Molecular Profiles of Nonalcoholic Fatty Liver Disease. *Sci. Rep.* **9**, 12541 (2019). https://doi.org/10.1038/s41598-019-48746-5.

[12] Edgar, R., Domrachev, M. & Lash, A. E. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Res.* **30**, 207–210 (2002). https://doi.org/10.1093/nar/30.1.207.

[13] Suppli, M. P. et al. Hepatic transcriptome signatures in patients with varying degrees of nonalcoholic fatty liver disease compared with healthy normal-weight individuals. *Am. J. Physiol. Gastrointest. Liver Physiol.* **316**, G462–G472 (2019). https://doi.org/10.1152/ajpgi.00358.2018.

[14] Pantano, L. et al. Molecular characterization and cell type composition deconvolution of fibrosis in NAFLD. *Sci. Rep.* **11**, 18045 (2021). https://doi.org/10.1038/s41598-021-96966-5.

[15] Kozumi, K. et al. Transcriptomics Identify Thrombospondin-2 as a Biomarker for NASH and Advanced Liver Fibrosis. *Hepatology* **74**, 2452–2466 (2021). https://doi.org/10.1002/hep.31995.

[16] Fujiwara, N. et al. Molecular signatures of long-term hepatocellular carcinoma risk in nonalcoholic fatty liver disease. *Sci. Transl. Med.* **14**, eabo4474 (2022). https://doi.org/10.1126/scitranslmed.abo4474.

[17] Verschuren, L. et al. Development of a novel non-invasive biomarker panel for hepatic fibrosis in MASLD. *Nat. Commun.* **15**, 4564 (2024). https://doi.org/10.1038/s41467-024-48956-0.

[18] Steinhauser, S. et al. The transcription factor ZNF469 regulates collagen production in liver fibrosis. *JCI Insight* **10**, e182232 (2025). https://doi.org/10.1172/jci.insight.182232.

[19] Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **15**, 550 (2014). https://doi.org/10.1186/s13059-014-0550-8.

[20] Benjamini, Y. & Hochberg, Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *J. R. Stat. Soc. B* **57**, 289–300 (1995). https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

[21] Liberzon, A. et al. The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst.* **1**, 417–425 (2015). https://doi.org/10.1016/j.cels.2015.12.004.

[22] Fang, Z., Liu, X. & Peltz, G. GSEApy: a comprehensive package for performing gene set enrichment analysis in Python. *Bioinformatics* **39**, btac757 (2023). https://doi.org/10.1093/bioinformatics/btac757.

[23] Ramachandran, P. et al. Resolving the fibrotic niche of human liver cirrhosis at single-cell level. *Nature* **575**, 512–518 (2019). https://doi.org/10.1038/s41586-019-1631-3.